

Kalman Filtering based Target Tracking for Multistatic Sensing in ISAC Systems

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Abstract—The advancement of Integrated Sensing and Communication (ISAC) has facilitated the adoption of target tracking technologies, yet most existing research focuses on waveform design rather than specific tracking algorithms within ISAC systems. To address this gap, this paper proposes an effective target tracking method based on Kalman filter (KF) algorithms within the ISAC framework. The method filters noisy radar measurements, such as bistatic range (BR), bistatic range rate (BRR), and direction of arrival (DOA), to enhance tracking accuracy. Performance is assessed by using trajectory maps and root-mean-square error (RMSE) curves, demonstrating that the proposed method significantly improves estimation accuracy while increasing robustness to measurement noise, especially in DOA. Compared to our previously proposed geometric target localization method, this approach delivers superior tracking performance, making it highly suitable for real-time applications in ISAC systems.

Keywords—Integrated sensing and communication (ISAC), Kalman filter, multistatic sensing, target tracking.

I. INTRODUCTION

Integrated Sensing and Communication (ISAC) represents an emerging paradigm in wireless communication systems, designed to unify sensing and communication functionalities within a shared hardware platform [1]–[3]. By jointly leveraging spectral, spatial, and temporal resources, ISAC enables efficient utilization for both tasks [4]–[6]. This dual capability positions ISAC as a promising solution for critical applications, such as target tracking. Target tracking refers to the process of continuously tracking, localizing, and predicting the movement of a target over time, based on sensor observations [7], [8]. While much of the existing research emphasizes the use of communication waveforms to derive radar parameters, this paper shifts the focus towards the sensing functionality, specifically addressing target tracking within ISAC systems.

Kalman filters (KF) are widely regarded as one of the most popular methods for target tracking. In recent years, significant advancements have been made in KF-based tracking algorithms [8]–[15]. In this context, filtering refers

to the extraction of relevant information from noisy measurements during the target tracking process. For example, in [10], a KF-based method was proposed to achieve efficient tracking of multiple moving objects under challenging conditions. Similarly, KF has been employed for correcting predicted inertial navigation system (INS) trajectories using global positioning system (GPS) measurements [11]. Furthermore, in [12], an integrated KF approach for accurate ranging and tracking was introduced, utilizing phase-derived ranging and Bayesian inference in wideband radar to enhance both efficiency and precision. Beyond traditional KF applications, the work in [14] combined KF with a generalized regression neural network (GRNN), resulting in two tracking algorithms that demonstrated superior performance, achieving centimeter-level accuracy. Despite the effectiveness of these KF-based algorithms in various applications, few have addressed their application in ISAC systems, where communication transmitters/receivers serve as sensors, as opposed to conventional sensing methods like cameras, Lidar, or radar [13], [15]. To our knowledge, no existing methods apply KF to intermediate measurements such as bistatic range (BR), bistatic range rate (BRR), and direction of arrival (DOA). Incorporating KF to handle these intermediate noisy measurements could enhance certain localization algorithms that rely on such data but fail to adequately address measurement noise.

Building upon our previous work in target localization within ISAC systems, we are motivated to propose a novel target tracking method tailored for ISAC applications. In earlier research, as outlined in [6], [16], we developed ISAC systems utilizing the standardized 5G NR waveforms, demonstrating the feasibility of using a single waveform for both communication and sensing purposes. The systems successfully detected bistatic radar parameters, including BR, BRR, and DOA. Moreover, leveraging these parameters, our previous work in [17] proposed a multistatic sensing algorithm for localizing a moving target within the ISAC framework. In this paper, we extend this research by introducing an effective target tracking method, which integrates KF into the ISAC system. The contributions of this paper are summarized below:

- A novel target tracking framework leveraging multistatic sensing within ISAC systems.
- An effective target tracking method that first filters the noisy radar parameters, such as BR, BRR, and DOA, obtained from the ISAC receiver. These filtered

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This research is supported by the National Research Foundation, Singapore and Infocomm Media Development Authority under its Future Communications Research Development Programme

measurements are then used to estimate the target's trajectory more accurately.

The remainder of this paper is organized as follows. In Section II, we formulate the tracking problem of a moving target in an ISAC system and briefly demonstrate the localization algorithm. The proposed KF-based tracking algorithm is presented in Section III. Section IV gives the simulation results and validates the performance. Finally, some conclusions are drawn in Section V.

II. SYSTEM MODEL

In this section, the system model is presented. As shown in Fig. 1, we consider a 2-D ISAC system model, where there is user equipment (UE) acting as transmitters (Tx) to transmit uplink ISAC signal to one base station as a receiver (Rx) which can detect the BR, BRR, and DOA. In this system, the uplink signal can be reflected from the target to Rx. The i -th Tx is located at (L_i, θ_i) in polar coordinates. The Rx is located at the origin, $(0, 0)$. The unknown position of the target is (L_0, θ_0) and its velocity is (v_0, φ_0) . In this system, the detected BR, BRR, and DOA can be expressed as

$$\Delta_i = L_0 + b_i - L_i + n_i, \quad (1)$$

$$\dot{\Delta}_i = \dot{L}_0 + \dot{b}_i - \dot{L}_i + \dot{n}_i, \quad (2)$$

$$\beta_i = \theta_0 + o_i, \quad (3)$$

where b_i is the distance between the target and the i -th Tx, $\dot{b}_i$ is the rate of change of b_i , n_i , $\dot{n}_i$, and o_i are additive zero-mean Gaussian noises with variances, σ_1^2 , σ_2^2 , and σ_3^2 , respectively.

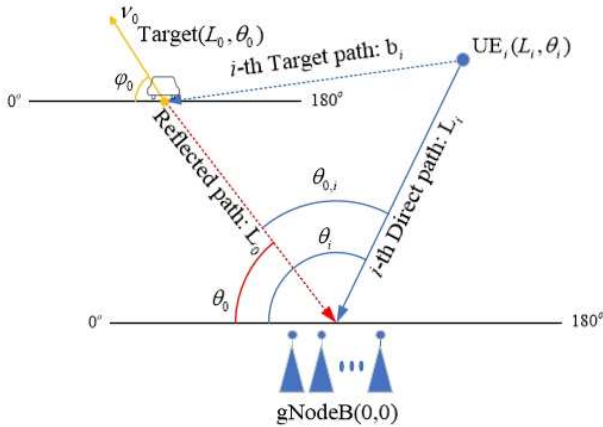


Fig. 1. System Model.

From geometric relation and our previous work in [17], we can get the estimated distance between the target and the Rx

$$\hat{L}_{0i} = \frac{\Delta_i^2 + 2\Delta_i L_i}{2\Delta_i + 2L_i - 2L_i \cos(\theta_i - \beta_i)}. \quad (4)$$

For N Tx and 1 Rx, we have N Tx-Rx pairs. The collection of noisy DOA measurements is $\theta = [\beta_1, \beta_2, \dots, \beta_N]^T$ and the collection of estimated ranges is $L = [\hat{L}_{01}, \hat{L}_{02}, \dots, \hat{L}_{0N}]^T$. Thus, we can obtain the estimated

location of the target $(\hat{L}_0, \hat{\theta}_0)$ through the maximum likelihood (ML) principle from the N pairs.

$$\hat{\theta}_0 = \arg \min_{0 \leq \theta_i \leq \pi} \|\theta_0 - \theta\|; \quad (5)$$

$$\hat{L}_0 = \arg \min_{0 \leq L_i \leq A} \|L_0 - L\|. \quad (6)$$

Similarly, from geometric relation and our previous work in [17], we can re-express the BBR with v_0 and φ_0

$$\begin{aligned} \hat{\Delta}_i = & -v_0 \cos(\varphi_0 - \pi - \hat{\theta}_0) \\ & + v_0 \cos(\varphi_0 - \theta_i + \text{sign}(\hat{\theta}_0 - \theta_i)(\hat{\alpha}_i - \pi)), \end{aligned} \quad (7)$$

where

$$\hat{\alpha}_i = \sin^{-1} \frac{\hat{L}_0 \sin(\hat{\theta}_0 - \theta_i)}{\sqrt{L_i^2 - 2L_i \hat{L}_0 \cos(\theta_i - \hat{\theta}_0) + \hat{L}_0^2}}.$$

Therefore, the ML estimation of (v_0, φ_0) is

$$(\hat{v}_0, \hat{\varphi}_0) = \arg \min_{0 \leq v_0 \leq V, 0 \leq \varphi_0 \leq \pi} \|\hat{\Delta} - \dot{\Delta}\|, \quad (8)$$

where $\dot{\Delta} = [\dot{\Delta}_1, \dot{\Delta}_2, \dots, \dot{\Delta}_N]^T$ and $\hat{\Delta} = [\hat{\Delta}_1, \hat{\Delta}_2, \dots, \hat{\Delta}_N]^T$.

In summary, the position and velocity estimation of the target can be expressed as $(\hat{L}_0, \hat{\theta}_0, \hat{v}_0, \hat{\varphi}_0)$.

III. KALMAN FILTERING BASED TARGET TRACKING

In this section, a novel target tracking algorithm based on KF is presented. The target tracking process can be categorized into four stages. First, BR, BRR, and DOA are acquired through multistatic radar sensing in the ISAC receiver. Second, the BR and BRR measurements are refined using a KF. Third, the filtered BR, BRR, and DOA data are employed to estimate the target's position and velocity. Finally, the position and velocity are continuously tracked over time. The corresponding process flowchart is presented in Fig. 2. The effective filtering algorithm is the main and new algorithm in the proposed framework.

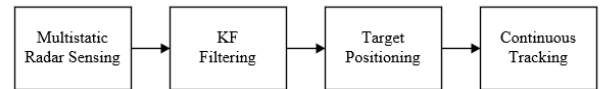


Fig. 2. Target Tracking Process

Considering the proposed system model, we measure BR and BRR at regular time intervals, and at the $(k-1)$ -th timestep, the BRR can be expressed as the discrete difference of the BR

$$\dot{\Delta}_{i,k-1} = \frac{\Delta_{i,k} - \Delta_{i,k-1}}{t_{i,k} - t_{i,k-1}}, \quad (9)$$

where $t_{i,k}$ is the moment at the k -th measurement.

Therefore, we can express the BR at the k -th timestep with BR and BRR at the previous timestep as

$$\Delta_{i,k} = \Delta_{i,k-1} + \dot{\Delta}_{i,k-1}(t_{i,k} - t_{i,k-1}). \quad (10)$$

Considering BR and BRR as two states of the target, to simplify, we write the state vector as $(x_{1,k}, x_{2,k})$. Then, we can express the state equations with process noises as

$$x_{1,k} = x_{1,k-1} + x_{2,k-1}(t_{i,k} - t_{i,k-1}) + \omega_{1,k-1}, \quad (11)$$

$$x_{2,k} = x_{2,k-1} + \omega_{2,k-1}, \quad (12)$$

where $x_{1,k}$ and $x_{2,k}$ are BR and BRR at the k -th timestep respectively, and $\omega_{1,k-1}$ and $\omega_{2,k-1}$ are the process noises of BR and BRR respectively, which satisfy zero-mean Gaussian distribution. Re-express the state equations in matrix format

$$\begin{bmatrix} x_{1,k} \\ x_{2,k} \end{bmatrix} = \begin{bmatrix} 1 & (t_{i,k} - t_{i,k-1}) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_{1,k-1} \\ x_{2,k-1} \end{bmatrix} + \begin{bmatrix} \omega_{1,k-1} \\ \omega_{2,k-1} \end{bmatrix}, \quad (13)$$

i.e.

$$\mathbf{X}_k = \mathbf{A}\mathbf{X}_{k-1} + \mathbf{\Omega}_{k-1}, \quad (14)$$

where $\mathbf{A}$ is the state transition matrix, $\mathbf{\Omega}_{k-1}$ is the system process noise which is zero-mean Gaussian white noise with covariance $\mathbf{Q}$.

Denoting the measurement vector including observed BR and BRR as $(z_{1,k}, z_{2,k})$, we express the measurement equations with measurement noise as

$$z_{1,k} = x_{1,k} + u_{1,k}; \quad (15)$$

$$z_{2,k} = x_{2,k} + u_{2,k}, \quad (16)$$

where $z_{1,k}$, $z_{2,k}$ are the measured BR and BRR at the k -th time step, and $u_{1,k}$, $u_{2,k}$ are the measurement noise, which is the zero-mean Gaussian white noise. Express measurement equations in matrix format

$$\begin{bmatrix} z_{1,k} \\ z_{2,k} \end{bmatrix} = \begin{bmatrix} x_{1,k} \\ x_{2,k} \end{bmatrix} + \begin{bmatrix} u_{1,k} \\ u_{2,k} \end{bmatrix}, \quad (17)$$

i.e.

$$\mathbf{Z}_k = \mathbf{X}_k + \mathbf{U}_k, \quad (18)$$

where $\mathbf{U}_k$ is the system process noise which is a zero-mean Gaussian white noise vector with covariance $\mathbf{P}$.

Therefore, the state at the k -th timestep relates to the previous measurement. Given the initial value of the state vector and noise covariance, we can make real-time estimates of the state by iteration.

There are two stages in the KF filtering process, prediction and correction. First, not knowing the noises ω_k and $\mathbf{V}_k$, we can make a priori estimation of the state as

$$\hat{\mathbf{X}}_k^- = \mathbf{A}\hat{\mathbf{X}}_{k-1}^-, \quad (19)$$

and the measurement as

$$\hat{\mathbf{X}}_{k, \text{measurement}} = \mathbf{Z}_k. \quad (20)$$

Estimate the noise by finding the difference between the priori estimation of the state and the measurement and

taking the weights to make a posteriori estimation of the state

$$\hat{\mathbf{X}}_k = \hat{\mathbf{X}}_k^- + \mathbf{K}_k(\mathbf{Z}_k - \hat{\mathbf{X}}_k^-), \quad (21)$$

where the coefficient $\mathbf{K}_k$ is the Kalman gain, which indicates whether we trust the estimate or the measurement more. Therefore, the task of KF is to find the best $\mathbf{K}_k$ at each timestep.

Assuming the prediction error $\mathbf{e}_k = \mathbf{X}_k - \hat{\mathbf{X}}_k^- \sim \mathcal{N}(0, \mathbf{R})$, then we have

$$\mathbf{R} = \mathbf{E}[\mathbf{e}\mathbf{e}^T] = \mathbf{E}[(\mathbf{X}_k - \hat{\mathbf{X}}_k^-)(\mathbf{X}_k - \hat{\mathbf{X}}_k^-)^T], \quad (22)$$

where $\mathbf{E}[\bullet]$ represents taking the expectation. To get the best Kalman gain, the trace of $\mathbf{R}$ should be minimized, and by solving this, we can get

$$\mathbf{K}_k = \frac{\mathbf{R}_k^-}{\mathbf{R}_k^- + \mathbf{P}}, \quad (23)$$

where

$$\mathbf{R}_k^- = \mathbf{E}[\mathbf{e}_k \mathbf{e}_k^T] = \mathbf{E}[(\mathbf{X}_k - \hat{\mathbf{X}}_k^-)(\mathbf{X}_k - \hat{\mathbf{X}}_k^-)^T] = \mathbf{A}\mathbf{R}_{k-1}\mathbf{A}^T + \mathbf{Q}.$$

Last, in the correction stage, the state and the covariance are updated at each timestep.

With the filtered BR and BRR, we can repeat the proposed target positioning in [17]. Reducing the errors in the original data, it can avoid introducing larger errors in the follow-up process.

IV. SIMULATIONS

In this section, we simulate an ISAC system composed of one gNodeB equipped with an antenna array as Rx and three UEs as TxS to track the position and velocity of a single target moving uniformly. The positions of the Rx and TxS are shown in Table I. This system can extract BR, BRR, and DOA with zero-mean Gaussian noises from the uplink ISAC signal, and the standard deviations are expressed as σ_1 , σ_2 , and σ_3 , respectively. The system takes measurements after every regular sampling interval of $(t_{i,k} - t_{i,k-1})$. These parameters are shown in Table II.

TABLE I. POSITIONS OF DEVICES IN THE SYSTEM

Devices	Polar	Cartesian
Rx	(0, 0)	(0, 0)
Tx ₁	(50, 0)	(-50, 0)
Tx ₂	(20, 45)	(-10 $\sqrt{2}$, 10 $\sqrt{2}$)
Tx ₃	(25, 135)	(12.5 $\sqrt{2}$, -12.5 $\sqrt{2}$)

TABLE II. SYSTEM PARAMETERS

Parameters	Values
Sampling Interval, $(t_{i,k} - t_{i,k-1})$	0.1 [s]
Standard Deviation of BR, σ_1	10 [m]
Standard Deviation of BRR, σ_2	1 [m/s]
Standard Deviation of DOA, σ_3	1 [°]

Initially, the target is located at (100, 100) m in the 2-D Cartesian coordinate and moving at a velocity of (10, 5) m/s. Assuming that the state of the target is unknown at the beginning, we set the initial value of the state vector to 0. With these parameters, the proposed target tracking algorithm is simulated within 10 seconds. Fig. 3 shows the comparison of the true trajectory, the trajectory calculated directly from measurements and the filtered trajectory in the trajectory map. According to the figure, the filtered trajectory is closer to the true trajectory than that before filtering, which demonstrates that the target tracking method has better tracking performance and can efficiently reduce the estimation error.

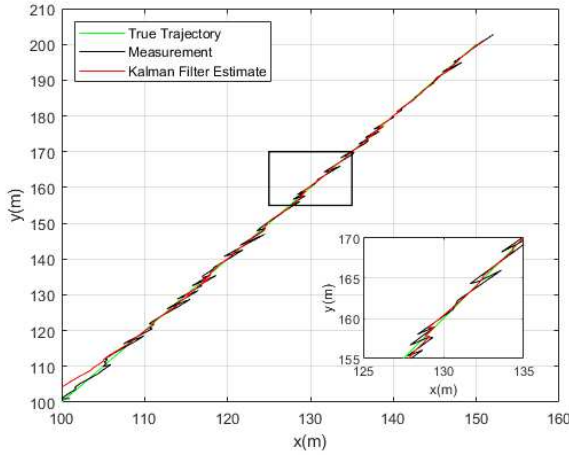


Fig.3. Trajectory map.

For quantitative analysis, we compute the RMSE averaged over M iterations, followed by a time-averaged computation of the results.

$$\text{RMSE}(\text{position}) = \frac{\sum_{t=t_0}^T \sqrt{\frac{\sum_{m=1}^M \left(\hat{L}_{0,m}^2 - 2L_0^o \hat{L}_{0,m} \cos(\hat{\theta}_{0,m} - \theta_0^o) + L_0^{o2} \right)}{M}}}{T - t_0}; \quad (24)$$

$$\text{RMSE}(\text{velocity}) = \frac{\sum_{t=t_0}^T \sqrt{\frac{\sum_{m=1}^M \left(\hat{v}_{0,m}^2 - 2v_0^o \hat{v}_{0,m} \cos(\hat{\phi}_{0,m} - \phi_0^o) + v_0^{o2} \right)}{M}}}{T - t_0}, \quad (25)$$

where $(\hat{L}_{0,m}, \hat{\theta}_{0,m})$ and $(\hat{v}_{0,m}, \hat{\phi}_{0,m})$ are the estimated position and velocity of the target at the m -th trial by target positioning, (L_0^o, θ_0^o) and (v_0^o, ϕ_0^o) are the true position and velocity of the target, t_0 represents the number of timesteps required for setting up the filtering process, while T denotes the total number of timesteps in the filtering process. The Monte Carlo count of the simulation is 5000. Setting $\sigma_3 = 0, 0.5^\circ$, and 1° respectively, we simulate RMSE curves over σ_1 varying from 10^{-1} to 10^1 m when keeping $\sigma_2 = 0.1$ m/s.

The RMSE curves for position and velocity are illustrated in Fig. 4 and Fig. 5, respectively. In Fig. 4, with a fixed σ_1 , an increase in σ_3 leads to a corresponding

increase in RMSE. When σ_1 is small, the differences between the three curves are significant; however, as σ_1 increases, these differences diminish. Consequently, as σ_1 continues to grow, the RMSE performance of the proposed method converges, irrespective of the value of σ_3 . In Fig. 5, the velocity RMSE also rises with increasing σ_1 and σ_3 . However, the curves remain nearly constant when σ_1 is less than 3 m. Additionally, as σ_3 varies from 0° to 1° , the proposed filtering method exhibits significantly improved performance compared to unfiltered results, with no angular errors in both position and velocity tracking.

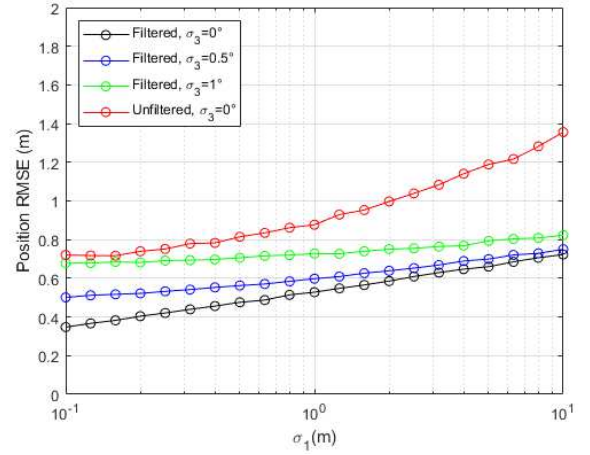


Fig. 4. Position RMSE comparison.

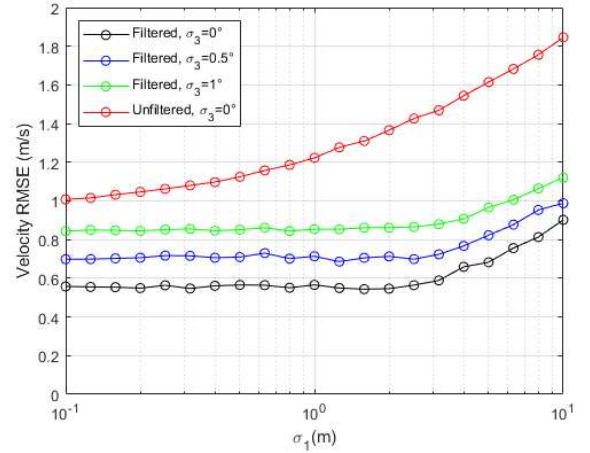


Fig. 5. Velocity RMSE comparison.

V. CONCLUSIONS

In this paper, we have presented an effective target tracking method for ISAC systems, utilizing KF to enhance tracking accuracy and robustness. By filtering noisy radar parameters such as BR, BRR, and DOA, the proposed approach significantly improves the estimation performance compared to the method in [17], when historical estimates are used for filtering and prediction. This work addresses the existing research gap in ISAC-specific tracking algorithms and provides a foundation for more accurate and reliable tracking in next-generation wireless networks. Future work will focus on extending this approach to more complex ISAC scenarios and further optimizing its computational efficiency for real-time applications.

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